

DRAFT RESEARCH PROJECT

Bitcoin Mathematics as the Map

Companies bridging traditional finance, law and the Bitcoin blockchain as the territory

The goal is capital migration: make it institutionally possible for money governed by traditional finance to move onto the Bitcoin blockchain.

CENTRAL QUESTION

Which mathematical guarantees does Bitcoin provide, which institutional facts do finance and law still require, and which companies already connect the two?

This paper is designed as a living market map. Company names are shown in bold throughout so market participants remain visible as the argument develops.

1. Executive thesis

**Bitcoin can settle value without asking law or finance for permission.
Large pools of traditional capital cannot move without custody,
identity, accounting, authority and legal remedies.**

The bridge is the market

The investment thesis is not that Bitcoin must reproduce traditional finance inside its protocol. It is that companies can connect Bitcoin's narrow, machine-verifiable guarantees to the wider institutional facts required by asset managers, banks, fiduciaries, auditors, courts and families.

The commercially important object is therefore not a purely on-chain solution. It is a controlled translation between two systems of truth:

Bitcoin evidence + institutional records = financeable Bitcoin

Here, **Bitcoin evidence** means signatures, scripts, transaction history and confirmed outputs. **Institutional records** means identity, account ownership, mandates, liabilities, approvals, legal title and recourse. **Financeable Bitcoin** means Bitcoin that an institution can lawfully buy, hold, audit, govern, transfer and pass to successors.

Research structure

- 2 The capital migration problem
- 3 What Bitcoin's mathematics provides
- 4 What traditional finance requires
- 5 What law requires
- 6 The bridge architecture
- 7-10 Existing company territory
- 11 Research conclusions and open questions
- 12 Market participant index and sources

2. The capital migration problem

Traditional capital begins inside a dense institutional system: bank accounts, brokerage accounts, funds, trusts, pensions, corporate treasuries and estates. Moving that capital into Bitcoin is not a single transaction. It is a chain of institutional permissions.

Stage	Traditional-finance question	Bitcoin-side requirement
1. Allocate	May this person or institution own Bitcoin?	Investment mandate and eligible product
2. Fund	How does fiat become BTC?	Exchange, broker, bank rail or OTC execution
3. Settle	Did the purchase reach the intended wallet?	Verified address, transaction and confirmations
4. Safeguard	Who can move it, and under what controls?	Keys, scripts, multisig or MPC policy
5. Account	How is it valued, reconciled and reported?	On-chain data joined to books and records
6. Govern	Who may approve transfers?	Role, mandate and transaction-policy controls
7. Exit or succeed	How is it sold, distributed or inherited?	Executable transfer plus valid legal authority

The migration equation

$$\text{Migratable capital} = \text{eligible capital} \times \text{trusted access} \times \text{operational control} \times \text{legal certainty}$$

Eligible capital is money permitted to buy Bitcoin. **Trusted access** is an accepted route from fiat or securities infrastructure into BTC. **Operational control** is a custody and authorization design that survives error, theft and staff changes. **Legal certainty** is evidence of ownership, authority, accounting treatment and remedies. Because the terms multiply, weakness in any one term can stop the whole migration.

This explains why bridge businesses can become valuable even when their cryptographic contribution appears small: they remove institutional bottlenecks from a much larger capital flow.

3. What Bitcoin's mathematics provides

Bitcoin supplies precise answers to narrow questions. Each formula below defines every term and separates what is proven from what must be supplied elsewhere.

SIGNATURE VALIDITY

$\text{Verify}(\text{pk}, \text{m}, \text{sig}) = 1$

pk is a public key; **m** is the exact message or transaction; **sig** is the signature; **1** means the verification rule accepts it.

This proves cryptographic validity, not legal authority.

SCRIPT SATISFACTION

$\text{Script}(\text{witness}, \text{output}) = \text{true}$

witness is the data supplied to spend; **output** contains the spending condition; **true** means Bitcoin's rules accept the condition as satisfied.

The script does not know why the spend occurred.

THRESHOLD CONTROL

$\text{valid signers} \geq t \text{ of } n$

n is the number of configured keys; **t** is the minimum required; $\geq$ means at least that many participants satisfy the policy.

This enables multisig resilience and shared control.

LEDGER STATE

$\text{Balance}(\text{U}, \text{h}) = \text{sum}(\text{value}(\text{u}))$

U is a selected set of unspent outputs; **h** is a block height; **u** is one output; **sum** adds their values.

This establishes a selected on-chain balance at a defined point.

HASH COMMITMENT

$\text{H}(\text{document}) = \text{digest}$

H is a cryptographic hash function; **document** is the input data; **digest** is its fixed-length fingerprint.

This detects alteration; it does not prove the document was true.

RESERVE RATIO

$R = A / L$

R is the reserve ratio; **A** is the verified asset set; **L** is the reported liability set; **/** means divided by.

The result is only as complete as A and L.

Mathematics proves that supplied evidence satisfies defined rules. It does not identify every relevant real-world fact.

4. What traditional finance requires

Traditional finance is not merely a slower payment network. It is a system for assigning identities, duties, claims, liabilities and remedies. Institutional Bitcoin adoption therefore requires controls beyond transaction validity.

Institutional need	Question it answers	Evidence normally used
Client identity	Who is the customer or beneficial owner?	KYC/KYB records, corporate documents, sanctions checks
Asset entitlement	Whose balance is this?	Account ledger, custody agreement, segregation records
Authorization	Who may instruct a transfer?	Mandates, roles, approval matrices, board resolutions
Safekeeping	Can one failure cause permanent loss?	Key architecture, recovery design, tested controls
Accounting	Do books reconcile to assets and liabilities?	Subledgers, confirmations, prices, reconciliations
Solvency	Are customer claims fully covered?	Proof of assets, liability population, encumbrance review
Liquidity and execution	Can size be acquired or sold efficiently?	Brokerage, OTC, exchange and settlement infrastructure
Auditability	Can an independent party reproduce the evidence?	Logs, source data, control reports, audit procedures
Recourse	What happens after error, fraud or insolvency?	Contracts, insurance, regulation and courts

The institutional control equation

$$\text{Approved transfer} = \text{valid Bitcoin spend} + \text{authenticated instruction} + \text{authorized role} + \text{policy compliance}$$

The plus signs mean all components are required in the institutional workflow. Bitcoin verifies the first component. Companies such as [Fireblocks](#) and [BitGo](#) commercialize the remaining authorization and operational layers.

5. What current law requires

Law asks who holds rights and duties, not merely which signature satisfies a script. A legally effective Bitcoin arrangement must connect technical control to legal persons, capacities and instruments.

Legal concept	Bitcoin evidence can show	Additional legal evidence required
Ownership	A script or key could authorize movement	Title, contract, trust, account or transaction records
Identity	A pseudonymous key or address	Verified person or legal entity
Capacity	Nothing about age, office or competence	Corporate role, trustee status, agency or guardianship
Authority	The spending condition was satisfied	Mandate, consent, fiduciary power or court order
Intent	Exact transaction data was signed	Absence of mistake, fraud, coercion or unauthorized use
Succession	Keys remain usable or unusable	Death, probate, trust terms and beneficiary entitlement
Remedy	A confirmed transaction remains in history	Injunction, restitution, damages or constructive trust

The core legal mismatch

Valid signature does not imply lawful authority

A valid signature can be produced by an owner, trustee, custodian, employee, thief or coerced signer. Bitcoin sees the same cryptographic result. Law assigns different consequences. The bridge is therefore documentary and institutional: identity records, governing instruments, custody contracts, approval logs and courts.

Regulated custody illustrates the bridge. [Anchorage Digital](#), [Coinbase Custody](#), [BitGo](#) and [Fidelity Digital Assets](#) combine key control with named accounts, contracts, audited controls and regulated entities.

6. The bridge architecture

A complete bridge is a stack. Different firms occupy different layers, and many successful companies span several.

Layer	Function	Representative companies
Access and execution	Convert fiat or securities capital into BTC and provide liquidity	Coinbase Prime , Kraken , River , NYDIG , Galaxy
Investment wrappers	Fit Bitcoin exposure into brokerage, advisory and fund mandates	BlackRock iShares , Fidelity , Grayscale , Bitwise , ARK/21Shares
Regulated custody	Bind controlled wallets to customer accounts and regulated safekeeping	Coinbase Custody , BitGo , Anchorage Digital , Fidelity Digital Assets , Komainu
Collaborative custody	Share control while preserving client-held keys and recovery paths	Unchained , Casa , Nunchuk , Theya , Keeper
Wallet governance	Apply roles, policies and approval workflows before signing	Fireblocks , Copper , Fordefi , BitGo
Data and attribution	Connect transaction graphs to counterparties and risk	Chainalysis , TRM Labs , Elliptic , Crystal
Reserve and audit evidence	Join on-chain assets to liabilities, books and attestations	River , Kraken , DACS , audit firms
Succession	Join key recovery to death, legal title and claim procedures	Casa , Unchained , trust companies, estate professionals

The bridge is not one company or one protocol. It is an institutional supply chain that ends in an on-chain UTXO and begins with capital governed by existing law.

7. Existing territory: custody and execution

Custody is the most developed bridge because institutions require identifiable account ownership, operational controls and a responsible counterparty before allocating capital.

Coinbase Custody Institutional custody within the broader Coinbase Prime stack. The bridge combines customer accounts, custody agreements, trading and settlement access.	BitGo Regulated custody plus multisig and MPC wallet infrastructure. Its architecture spans safekeeping, transaction policy and self-custody variants.
Anchorage Digital Federally chartered digital-asset banking and qualified custody positioning, with segregated custody, trading and settlement services.	Fidelity Digital Assets Institutional custody and execution built inside a traditional financial group, including audited operational controls and bank-charter infrastructure.
Komainu / Copper Institutional custody and settlement-oriented services, including infrastructure designed to reduce exchange counterparty exposure.	River / Kraken Brokerage or exchange access with public reserve-verification practices that connect customer claims to controlled assets.

Investment wrappers are a separate bridge

Spot Bitcoin funds from [BlackRock iShares](#), [Fidelity](#), [Grayscale](#), [Bitwise](#) and [ARK/21Shares](#) let traditional brokerage capital obtain economic exposure while the underlying BTC is held through institutional custody. The investor owns a security claim, not a Bitcoin key. Nevertheless, fund inflows can cause BTC to be acquired and settled on-chain.

Research implication: distinguish **capital moving into Bitcoin the asset** from **investors moving into direct Bitcoin ownership**. Both expand demand, but their legal and custody structures differ.

8. Existing territory: control, governance and recovery

The next bridge category translates organizational authority into controlled signing.

Company	Primary bridge	Mathematical anchor	Institutional layer
Fireblocks	Enterprise wallet and transaction governance	Threshold signing / MPC	Roles, approval policies, limits, destination controls
BitGo	Custodial and self-custodial wallet architectures	2-of-3 multisig and MPC	Policies, operators, recovery and regulated custody
Copper	Institutional custody and settlement infrastructure	Distributed key control	Trading workflow, settlement and institutional operations
Fordefi	Institutional wallet operations	MPC signing	Policy engine, transaction simulation and approvals
Unchained	Collaborative Bitcoin custody	2-of-3 multisig	Client-held keys, provider recovery key, inheritance support
Casa	Personal and family multi-key security	Multi-key vaults	Health checks, recovery and inheritance claim process
Nunchuk	Bitcoin multisig and collaborative custody tooling	Bitcoin multisig / PSBT	Coordination, assisted custody and recovery workflows
Theya / Keeper	Assisted Bitcoin custody	Multisig	User experience, key distribution and recovery assistance

The control equation

Institutional authorization = signing threshold + authenticated people + approved policy

A multisig or MPC threshold makes distributed control technically possible. It does not define employment roles, fiduciary duties or transaction purposes. The commercial products join the cryptographic threshold to organizational identity and policy.

9. Existing territory: reserves, accounting and intelligence

This territory turns public blockchain data into evidence that traditional finance can use.

River

Publishes proof of assets and proof of liabilities. Customers can compare controlled BTC with reported customer claims. This joins an observable asset set to a private account ledger.

Kraken

Uses proof-of-reserves reviews and customer inclusion verification across covered assets. The process illustrates both the value and scope limits of reserve evidence.

Chainalysis

Combines transaction graphs with ground-truth attribution, heuristics and external evidence to support compliance and investigations.

TRM Labs / Elliptic / Crystal

Competing intelligence platforms connecting on-chain activity to entities, typologies, sanctions and risk assessments.

DACS

Digital-asset attestation and audit-oriented evidence, including procedures aimed at professional reporting use cases.

Audit and accounting firms

Professional auditors test ownership, controls, completeness, valuation and reporting using more than a signature or explorer screenshot.

Three different evidence claims

Claim	What supports it	Main limitation
Existence	Identified outputs in the ledger	The selected set may be incomplete
Control	Signature, test transaction or observed ceremony	Does not alone prove exclusive or lawful control
Ownership and reporting	Contracts, account records, controls and audit evidence	Depends on external records and professional judgment

10. Existing territory: succession and legal continuity

Succession is where Bitcoin's irreversibility meets the long time horizon of wealth. The bridge must preserve security during life while making lawful access possible after death or incapacity.

Succession-ready Bitcoin = legal title + discoverable plan + sufficient keys + executable claim process

Legal title identifies the beneficiary and fiduciary. A **discoverable plan** ensures the right people know an arrangement exists without exposing secrets prematurely. **Sufficient keys** means the required signing threshold can eventually be reached. An **executable claim process** defines evidence, waiting periods, verification and transfer assistance.

Unchained	Casa
Combines 2-of-3 collaborative custody with inheritance documentation and operational support. Its materials explicitly separate transfer of title from transfer of key possession.	Combines multi-key vaults, recipient setup, health checks and a delayed access-request process. It operationalizes a succession path without treating Bitcoin as a death oracle.
Estate-planning professionals	Trust companies and custodians
Wills, trusts, beneficiary rights, fiduciary appointment and tax planning establish the legal side of succession.	Institutional account records, identity verification and fiduciary procedures can connect a court-recognized successor to controlled assets.

Research implication

The strongest succession products do not try to make Bitcoin detect death. They coordinate independent legal and technical processes. This is a general model for the entire bridge market: preserve Bitcoin's native guarantees while adding external evidence only where institutions need it.

11. Conclusions and open research questions

The existing market supports five initial conclusions.

Conclusion	Meaning
1. The bridge is already a major market	Custody, ETFs, brokerage, wallet infrastructure and analytics already move or support substantial capital.
2. No single bridge model dominates	Qualified custody, collaborative custody, multisig, MPC and securities wrappers solve different institutional constraints.
3. The most valuable firms bind two evidence systems	They combine on-chain facts with identity, policy, liabilities, contracts or professional assurance.
4. Capital exposure and direct ownership must be separated	A fund share can drive on-chain BTC acquisition without giving the investor keys or direct title to specific outputs.
5. Law is a complement, not a protocol defect	Legal identity, authority and remedies can remain external while Bitcoin provides settlement and verifiable state.

Open questions for the next revision

- Which bridge categories capture the greatest economics as traditional capital grows?
- Where do regulated custody and collaborative custody converge or remain structurally distinct?
- Which institutional controls can become independently verifiable without exposing wallet privacy?
- How should reserve evidence address liability completeness, encumbrances and snapshot risk?
- Which jurisdictions provide the clearest treatment of Bitcoin custody, property rights and fiduciary duties?
- Which companies are becoming systems of record rather than point solutions?
- How much capital reaches on-chain BTC through funds versus direct account or self-custody ownership?

Working thesis: the durable winners will make Bitcoin legible to institutions without weakening the properties that make Bitcoin valuable.

12. Market participant index and research sources

This is a representative living index, not a claim of exhaustive coverage. Names are grouped by the role emphasized in this draft; several companies span multiple categories.

Category	Named market participants
Access, trading and execution	Coinbase Prime , Kraken , River , NYDIG , Galaxy
Investment wrappers	BlackRock iShares , Fidelity , Grayscale , Bitwise , ARK/21Shares
Regulated / institutional custody	Coinbase Custody , BitGo , Anchorage Digital , Fidelity Digital Assets , Komainu , Copper
Collaborative / assisted custody	Unchained , Casa , Nunchuk , Theya , Keeper
Wallet and policy infrastructure	Fireblocks , BitGo , Copper , Fordefi
Blockchain intelligence	Chainalysis , TRM Labs , Elliptic , Crystal
Reserve / attestation examples	River , Kraken , DACs
Succession examples	Casa , Unchained , estate lawyers, trust companies and fiduciaries

Primary and professional sources used

Bitcoin technical specifications: [BIP-322 Generic Signed Message Format](#); [Bitcoin Developer Guide - Transactions](#); relevant Bitcoin Improvement Proposals on witness and script validation.

Company materials: official custody, wallet, inheritance, reserve and security documentation from [BitGo](#), [Unchained](#), [Casa](#), [Fireblocks](#), [River](#), [Kraken](#), [Anchorage Digital](#), [Fidelity Digital Assets](#) and [Chainalysis](#).

Professional and regulatory materials: [ICAEW Considerations for Auditing Cryptocurrencies](#); the 2025 joint U.S. banking-agency statement on crypto-asset safekeeping; OCC interpretations concerning crypto custody and fiduciary activity.

Next: validate every classification, add jurisdictions and economics, and maintain a dated market-change log.